

EFFICACY AND SAFETY OF ORAL CITICOLINE IN ACUTE ISCHEMIC STROKE: DRUG SURVEILLANCE STUDY IN 4,191 CASES

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SUMMARY

Citicoline is an essential precursor in the synthesis of phosphatidylcholine, a key cell membrane phospholipid, and is known to have neuroprotective effects in acute ischemic stroke. The aim of this study was to determine the efficacy and safety of oral citicoline in Korean patients with acute ischemic stroke. A drug surveillance study was carried out in 4,191 patients with a diagnosis of acute ischemic stroke. Oral citicoline (500-4000 mg/day) was administered within less than 24 h after acute ischemic stroke in 3,736 patients (early group) and later than 24 h after acute ischemic stroke in 455 patients (late group) for at least 6 weeks. For efficacy assessment, primary outcomes were patients' scores obtained with a short form of the National Institutes of Health Stroke Scale (s-NIHSS), a short form of the Barthel Index of activities of daily living (s-BI) and a modified Rankin Scale (mRS) at enrollment, after 6 weeks and at the end of therapy for those patients with extended treatment. All adverse reactions were monitored during the study period for safety assessment. All measured outcomes, including s-NIHSS, s-BI and mRS, were improved after 6 weeks of therapy ($P < 0.05$). Further improvement was observed in 125 patients who continued citicoline therapy for more than 12 weeks when compared with those who ended therapy at week 6. Improvements were more significant in the higher dose group (≥ 2000 mg/day) ($P < 0.001$). s-BI scores showed no differences between the early and late groups at the end of therapy. Citicoline safety was excellent; 37 side effects were observed in 31 patients (0.73%). The most frequent findings were nervous system-related symptoms (8 of 37, 21.62%), followed by gastrointestinal symptoms (5 of 37, 13.5%). Oral citicoline improved neurological, functional and global outcomes in patients with acute ischemic stroke without significant safety concerns.

Key words: Oral citicoline - Ischemic stroke - Efficacy - Safety - Somazina®

INTRODUCTION

Citicoline (cytidine diphosphate choline or cytidine 5'-diphosphocholine, Somazina®) is an essential precursor in the synthesis of phosphatidylcholine, a key membrane phospholipid (1, 2). Citicoline is normally present in all cells. Exogenous citicoline easily crosses the blood-brain barrier, and its administration for extended periods of time increases the phosphatidylcholine concentration in the brain (2). Citicoline inhibits neuronal phospholipid membrane breakdown and repairs the neuronal membrane after neuronal injury, ischemia and hypoxia—benefits for which it is well known. It also potentiates acetylcholine production and reduces free fatty acid accumulation in ischemic tissues (1, 3).

In experimental animals, exogenous citicoline proved its benefits by means of reducing cell membrane breakdown, decreasing free fatty acid levels and also reducing infarct volume (4, 5). Trials carried out in humans with various neurological diseases (6), including dementia (3, 7), trials in rats with traumatic brain injury (8), peripheral nerve injury (9) and spinal cord injury (10), and trials in humans with optic

neuropathy (11) demonstrated citicoline's clinical benefits. Previous trials in acute stroke patients demonstrated some significant improvement on secondary outcomes and post hoc analysis (12-15). A pooled analysis showed the efficacy of the drug in the treatment of patients with moderate to severe stroke (15).

Oral citicoline is water-soluble and has an oral bioavailability of over 90%, so the dose is no different for oral and parenteral forms of the drug (1). Based on the aforementioned trial results, citicoline has been used worldwide since the 1990s; however, its safety profile has not been reported in the Asian population. The aim of this study was to determine the efficacy and safety of oral citicoline in patients with acute ischemic stroke in Korea.

METHODS

Patients

Between March 2006 and March 2008, men and women aged ≥ 18 years at neurology, neurosurgery and rehabilitation departments of